

ONGOLE, India

WMO: 432210

Lat: 15.500N

Lon: 80.083E

Elev: 39

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	69.3	70.4	54.9	64.1	82.7	58.2	72.6	83.1	5.6	83.6	5.4	83.4	1.4	N/A	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	16.5	106.1	78.5	102.9	78.8	100.2	78.7	83.4	94.8	82.6	93.1	81.9	91.9	3.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.9	88.4	80.1	156.3	87.9	79.5	153.1	87.6	47.5	95.0	46.6	93.7	45.8	91.9	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.2	5.4	66.7	112.5	1.2	2.5	65.9	114.3	65.2	115.7	64.5	117.1	63.7	118.9		
			59.4	85.6	2.2	1.2	57.8	86.5	56.5	87.2	55.3	87.9	53.7	88.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.9	78.7	81.1	85.7	89.5	93.8	91.9	88.6	87.7	86.8	85.8	81.8	79.2
	DBStd	5.58	1.61	1.90	2.08	2.19	4.36	4.50	3.96	3.55	3.06	3.32	2.55	1.97
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	13109	889	872	1108	1185	1357	1258	1197	1169	1104	1110	953	907
	CDD65	7634	424	452	643	735	892	808	732	704	654	645	503	442
	CDH74	93815	2704	3737	7470	10074	13482	12141	10379	9536	8549	7949	4717	3077
	CDH80	49497	713	1309	3341	5762	9048	7853	5988	5170	4324	3735	1449	805
Wind	WSAvg	2.0	1.4	1.5	2.1	2.5	2.6	2.7	2.7	2.4	1.9	1.4	1.6	1.5
Precipitation	PrecAvg	48.7	0.3	0.4	0.5	0.8	2.7	6.0	9.1	9.0	8.2	8.3	2.7	0.8
	PrecMax	73.9	1.5	2.0	5.2	4.3	15.4	11.7	13.8	15.3	16.6	15.6	8.1	4.3
	PrecMin	35.3	0.0	0.0	0.0	0.1	0.5	2.4	5.4	5.0	2.7	0.9	0.0	0.0
	PrecStd	8.5	0.4	0.6	1.0	0.8	3.1	2.6	2.2	2.1	3.2	4.2	2.4	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.7	94.0	98.1	102.4	110.8	107.9	102.2	100.3	98.4	97.7	93.3	89.9
		MCWB	71.9	68.6	77.2	80.6	78.0	78.1	77.8	78.8	77.8	75.3	75.2	72.1
	2%	DB	86.9	90.6	95.1	99.0	107.8	104.8	99.7	98.2	96.3	95.4	90.6	87.7
		MCWB	72.0	72.4	76.5	80.4	78.0	78.2	77.3	77.7	77.7	76.1	75.2	73.2
	5%	DB	85.5	88.6	93.3	97.1	104.8	102.1	97.8	96.2	94.5	93.7	88.6	86.1
		MCWB	72.0	73.1	76.5	80.1	78.8	78.3	77.4	77.8	77.9	76.8	75.2	73.2
	10%	DB	84.1	86.9	91.6	95.4	101.4	99.5	95.7	94.2	92.6	91.8	86.8	84.5
		MCWB	71.7	73.1	76.3	79.9	79.6	78.0	77.1	77.6	78.2	77.3	75.4	72.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	77.7	81.0	83.4	84.8	83.9	81.4	81.8	82.1	81.7	79.7	77.6
		MCDB	83.5	86.7	91.5	95.0	96.0	95.0	89.4	90.4	90.5	89.3	85.7	83.6
	2%	WB	75.3	76.7	80.0	82.5	83.7	82.7	80.4	80.8	81.2	80.7	78.9	76.4
		MCDB	81.4	85.0	89.7	93.3	95.6	92.1	89.2	89.4	89.2	87.8	84.9	82.1
	5%	WB	74.3	75.9	79.2	81.8	83.1	81.7	79.7	80.1	80.6	80.0	78.2	75.6
		MCDB	80.8	84.0	88.1	91.9	94.4	91.3	89.1	88.9	88.3	87.2	84.0	81.5
	10%	WB	73.4	75.0	78.6	81.2	82.4	80.8	78.9	79.5	79.9	79.4	77.6	74.8
		MCDB	80.4	82.9	87.0	90.8	93.0	91.1	89.1	88.6	87.7	86.8	83.1	80.8
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.3	15.0	14.4	16.5	14.3	12.2	11.9	11.5	12.0	11.6	13.5
		MCDBR	15.2	16.5	16.2	15.7	20.5	16.8	14.7	14.0	13.8	14.7	13.8	14.7
		MCWBR	4.0	4.3	3.8	3.9	5.6	4.6	3.8	3.6	4.0	4.2	3.9	4.5
	5% WB	MCDBR	12.7	13.7	14.4	14.9	17.0	14.9	13.5	12.9	12.2	11.7	10.2	11.6
MCWBR		3.9	4.2	3.8	4.0	5.2	5.4	4.4	4.4	4.4	4.4	3.8	4.2	
Clear-Sky Solar Irradiance	taub	0.507	0.496	0.548	0.568	0.638	0.585	0.554	0.540	0.536	0.561	0.540	0.561	
	taud	2.008	2.061	1.946	1.936	1.790	1.931	2.028	2.072	2.080	1.996	1.993	1.900	
	Ebn at Noon	248	259	249	243	223	233	241	247	248	238	237	228	
	Edh at Noon	54	53	61	62	71	61	55	53	53	56	54	59	
All-Sky Solar Radiation	RadAvg	1536	1819	1996	2118	1979	1622	1474	1514	1544	1473	1408	1393	
	RadStd	58	74	71	69	62	129	156	90	117	132	108	91	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.50	+0.65	N/A	N/A	+0.49	+0.45	N/A	N/A	N/A	N/A

Nomenclature: See separate page